
PARTNER NETWORKS: LEVERAGING EXTERNAL TIES TO IMPROVE ENTREPRENEURIAL PERFORMANCE

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EXECUTIVE SUMMARY

This paper examines the conditions under which successful partnership networks were formed by four entrepreneurial companies. Seven alliance partnerships were studied. Both sides benefited through product advances, administrative process improvements, and rapid response times due to the greater levels of information exchange and coordination through computers. Key to understanding the partnerships was the development of trust between organizations. The value of these partnerships in terms of benefits to smaller companies has not received adequate attention in the literature. Yet forging such collaborative alliances seems crucial in explaining the ability of smaller firms to grow and to innovate.

These networks should be seen as a competitive alternative to vertical integration. Smaller-scale entrepreneurial firms lack the financial resources to vertically integrate steps in the value added chain. This research suggests that a network organizational form can be cultivated by smaller companies to realize the benefits of vertically integrated functions while avoiding the bureaucratic inefficiencies of that organizational form. The network strategy of building close collaborative alliances with a limited set of suppliers and customers enables a firm to stabilize itself while remaining flexible and responsive to a changing market. An important aspect of strategic planning for the entrepreneurial firm is to identify prospective partners and consciously initiate and build partnerships with responsive firms. The data gathered indicate that these alliances do not form by chance but can be studied as patterned, predictable exchange structures that can be replicated and used to improve a firm's competitive position against larger players.

The paper argues for an expansion of our concept of entrepreneurship to include the effective management of partnership networks. We should also expand our ideas about organizational forms to recognize the network structure as an effective governance arrangement for entrepreneurial com-

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panies. The findings strongly suggest that entrepreneurial firm's ability to identify, cultivate, and manage these network partnerships is critical to survival and success.

INTRODUCTION

Entrepreneurial researchers have looked at networks used by entrepreneurs at the initial stages of building a company (Birley 1985; Aldrich and Zimmer 1986). Others have discussed the potential for new organizational partnership combinations to contribute to growth and innovation (Stevenson 1983; Kanter 1989; Jarillo 1988; Lorenzoni and Ornatì 1988, Johnston and Lawrence 1988), but these accounts do not provide detailed analysis of how and why partnerships form, nor do they fully explore the benefits derived. This paper conducts such an inquiry and takes the next step to explore the conceptual implications of these relationships for our thinking about, and teaching about, organizations and entrepreneurship.

METHOD

The project was an ethnographic study. The data were gathered in a qualitative study of the processes by which high-growth smaller firms formed outside alliances. The analysis of process required in-depth examination of a limited number of examples, a design that has obvious limits for generalizations. The value of exploratory ethnographic research lies instead in its capacity to provide insights through rich detail and to generate hypotheses for further testing.

While the research design was not comparative at the outset, it should be pointed out that, to a degree, the comparative method was used throughout this research, in that interview respondents regularly compared these unusual cooperative alliances with more typical interfirm relationships. Another comparison made often by respondents juxtaposed these partnerships with a seemingly logical alternative arrangement; that is, vertical integration under single ownership.

This study was based on field interviews conducted by the author between June 1986 and July 1987. These interviews formed the core of a research project designed to look in detail at how cooperative partnerships formed between a set of entrepreneurial firms and a limited number of their suppliers and customers.¹ Over 50 interviews were conducted with individuals in the firms to discover the conditions under which this set of close exchange relations formed and what their impact on performance was.

The data were collected on high growth firms that met the following criteria:

- Rapid growth between 1980 and 1986
- 1985 revenues of at least \$10 million
- Sales growth of 20% or more compounded annually for the last five years

A thorough study required a small sample of entrepreneurial firms and a manageable number of their alliances. The decision was made to focus on four firms and seven alliances.

¹The focus on cooperative partnerships resulted from exploratory interviews conducted at 16 small high-growth firms. These firms were selected based on their rapid growth between 1980 and 1986. This set of interviews revealed that every entrepreneurial company relied heavily on a small number of unusually close outside relationships with other firms for its financial success, rapid growth, and innovativeness. The question for this project was posed: what was the nature of these relationships?

The four entrepreneurial firms represented a small but diverse sample. They included two manufacturing firms and two marketing/distribution firms. The products in the alliances included both high- and low-technology products. Certain products were commodity items and others were custom-made products. The firms' sizes in 1987 ranged from \$18 million in sales at the low end to \$250 million in sales for the largest company.

Members of management in the four firms identified at least one and usually two or three outside alliances that were maintained during the company's high-growth period and that had contributed to their competitive success. From this set of alliances seven were selected based on their diversity in size and function.² Note that whereas the companies were the initial unit of analysis, once it was decided to explore the network arrangements between firms, the dyadic relationships became a new unit of analysis. It should be kept in mind that these individual dyads were studied, not for their own sake, but to gain an appreciation of the nature and power of the network structures of which they were a critical constituent part. By studying the dyads one is able to see how they functioned and the extent to which these kinds of networked relations, as a system, significantly shape the entrepreneurial organizations involved.

A two-step field guide questionnaire was used that directed and structured the otherwise open-ended interviews. The first section of the questionnaire guide was a set of factual questions designed to collect information on each company's history, products, and markets. The second section focused on the network ties. The origins, the importance of the relationships to each side, the contract terms, the nature of routine interactions, and the benefits were explored.

FINDINGS

To summarize the initial findings, the research revealed that each entrepreneurial firm was a hub organization with a small number of stable exchange relationships (a network) that they maintained with favored outside companies. These ties contrasted sharply with the hub firms' more typical arm's-length market relationships, both in terms of their duration and the quality of interfirm communication. In effect, the entrepreneurial organizations had formed small, but dense, networks of ties with a handful of outside firms providing critical resources. These exchange relations more closely resembled well-managed vertically integrated functions than they did market-mediated exchanges. An impressive array of benefits to the entrepreneurial ventures resulted from these alliances. The detailed account of these benefits follows this discussion. But first we turn our attention to the conditions and processes that made these benefits possible.

²These alliances linked the entrepreneurial firms to other companies that were smaller, of equivalent size, or much larger than themselves. In most cases the primary economic transaction was the sale of a product; however, one exchange was a subcontracting arrangement for labor only. Another was an unusual relationship with a sales organization. Some firms were tightly interlocked; others appeared to be more loosely coupled. Some had frequent, even daily, contact; others communicated less often. The common features of this group of seven alliances were that each had lasted for at least five years and that each was seen by management as contributing significantly to its respective entrepreneurial firm's rapid growth and economic success. Interviews with alliance partners were conducted at the respective companies, with follow-up questions explored through a combination of face-to-face interviews and follow-up telephone conversations. The data collection included 54 interviews, averaging one and a half hours each. Interviews were conducted with individuals on both sides of the alliance. A special effort was made to locate and talk with both individuals originally involved in the formation of the alliances and persons still actively managing the relationships.

A. Process, Stage One: The Trial Period

A primary economic exchange (e.g., product for a price) provided the impetus for each tie. My informants described the evolution of the relationships in two stages. Initially the firms were engaged in arm's-length market relations. Like typical market exchanges, the transactions were price based, with limited history or future. The firms then moved to a start-up phase in which they saw themselves experimenting and attempting to build a relationship, not knowing whether the transaction arrangement would prove long- or short-term.

This period of time constituted a trial period that allowed prospective partners to evaluate each other and learn about their respective businesses. During this time, ground rules for routine interactions were set down and firms established their credibility both through demonstrated performance capabilities and their style of conducting business (e.g., fairness, straightforwardness). Explicit and implicit ground rules were established during this trial phase as the individuals and organizations moved incrementally toward set patterns of interaction and exchange.

The trial period required an accumulation of exchanges and an evaluation of the partner's trustworthiness. A marketing manager at one company described the trial period:

You can't start out with a full-blown relationship. It's got to be incremental. The other side takes a small step and you take a small step. There's gradual movement to get closer and closer. I started calling on Support Products in 1978 and they had about 300 people. Now they have 2000. When I first started I didn't know whether I could believe everything they told me. But experience, facts you collect, the transactions you have with different people tells you whether you can believe everything they say or not.

The trial period required a commitment of time to the development of exchange-specific patterns, what one informant called "the systems and procedures."

You go through a trial and error period. You learn from mistakes and they learn from our people. Incrementally, we become part of their operation. Every customer is unique and it takes time to get accustomed to their idiosyncracies. It takes about a year—a minimum of one year. Get up to the second year and you are rounding off the rough edges. Then you really get to hit the peaks of a truly efficient operation.

During the trial period, explicit and implicit rules began to take shape. Through the steady exchange of information, communications became routinized and procedures were agreed upon. The sales manager at a computer-hardware company talked about the increasing ease of communications as this learning period progressed:

It gets easier because you learn how each side wants to play the game and what they expect. So you learn what the unwritten rules are. Over a period of time you've gone through certain situations and you know how they are going to play it; and they know how you are going to play it. You have a history of situations, compromises, and solutions. With that background it makes it increasingly easier to do business.

It is interesting to note that formal contracts were not the driving force behind these alliances. Although four of the ties had written agreements, the participants agreed that written contracts were unimportant to the total picture. More important than formal contracts was trust. When asked what protected them from opportunistic behavior by the other side, the interviewers emphasized, time and again, trust. One individual remarked:

The contracts aren't worth the paper they are printed on. Because the relationship, the day-to-day operating relationship, is not managed by the verbiage contained in the contract. Contracts are used more as a safety net in case the people relationship falls apart.

With the repetition of incremental quid pro quo behavior, by the end of the trial period an implicit contract had developed based on trust, and the reciprocity in actions that trust encourages. Trust had emerged as a central component of the exchange. The vice president of the high-growth computer-hardware company described that firm's relationship with a key subcontractor:

It is like working with your own factory. There is full trust. When we call to say don't worry about cost, they know what we mean. They trust us to pay and we trust them to give us a reasonable price.

Trust referred to several aspects of behavior: confidence that the other side could be relied upon (for performance and for protecting proprietary information, for example); confidence that the relationship would not be exploited by the other side; confidence that extra effort would be consistently made; and, in return, that a partner gives the other side time and opportunity to adjust to changed circumstances rather than move abruptly to an alternative supplier. The data suggest that trust is achieved after a series of reciprocated exchanges that signal the firms' adherence to a shared interpretation of fairness. Once firms reach what can be viewed as a trust threshold (a common understanding of fairness and the assumption of trust), short-term losses are set against long-term gains; and each side begins to play for future as well as present stakes. Once trust was established as a guiding foundation of exchange, cooperation became the operating mode. This mode replaced the adversarial relationship described by interviewers as the more typical way of doing business with their other suppliers or customers.

B. Process, Stage Two: The Partnership

If the efforts to coordinate the firms' activities around their mutual interest were successful, the trial period shifted to a second phase of routine operation that was seen as a structurally complete, organizationally integrated, and "good" relationship. This second stage was easily distinguished from the first because it required the integration of the two firms, investment processes, and a commitment to continuous improvement through reciprocated exchanges; as opposed to the initial trial-and-error building processes typical of Stage One.

Stage Two was characterized by the strong commitment to the preservation of the partnership and the integration of the partnered firms through extensive and frequent communications.

You have to have relationships with engineering and purchasing, a breadth of relationships. The key is not to give them any reason to look elsewhere—which means you have to catch problems immediately. That requires a great deal of communications across all levels of both organizations.

The nature and extent of communications in these alliances caused informants to remark repeatedly that they resembled well-coordinated, vertically integrated units. The firms became tightly integrated through frequent contact across organizational and functional boundaries.

We feel we would not communicate with them any differently if we were attached to the back of their building. We have different steps in the ladder that communicate. Everybody has their contact at their organization as they would if we were attached to the building. There is constant contact. We are part of their operation.

The mature partnership also involved investments, not just of time and energy, but of capital investments as well. One division manager of the large printing company said this:

We have a significant investment in [the entrepreneurial firm's] business and they know it. We have people and equipment allocated to them. Some of the equipment is unique.

The manufacturing vice president of the computer manufacturing firm described the partnerships with his select suppliers in this way:

It is up to us to lessen the impact of constant changes, to make it easier for them because making it easier for them brings benefits back to us. It means investing time. It comes down to allocating resources and getting information to them quickly so that they can better service us.

Investments were expected to be returned, although the particular form and time frame for reciprocity was unspecified. The planning director of a telephone equipment company talked about this aspect of the exchange process:

It is like a balance; it is like a scale. In return for the commitment on their part we say we are committed to you and we prove it . . . So it's a quid pro quo. It's a balanced relationship that says you make investments, we make investments; you take risks, we take risks; you perform, we perform. That's the basis on which you build trust and everything that I would consider to be a strong successful relationship.

This reciprocity extended to the commitment each side made to enhance and expand the exchange process. This aspect was a dynamic element that I called *kaizen orientation*.³ Kaizen is a Japanese term that refers to constant improvement in any given situation. Constant improvement was the result of experimentation, risk taking, innovation, and a focus on quick response and problem solving. These quotations describing three partnerships provided typical descriptions:

It takes a lot of work to maintain the relationship, including day-to-day effort and constantly improving service.

You could call it positive maintenance—making sure everything that we have is going well. Review that and make sure everybody is up to speed on where we are going and that there aren't any changes or surprises coming down the pike.

Even though there is a production schedule the relationship has to be attended to. You call them virtually every day to ask, "How can I help?"

In summary, Stage Two was characterized by extensive information-sharing through dense communication channels between the two units. The incremental, small-scale exchanges that characterized Stage One were replaced with a pattern of reciprocated investments of time, and in people and equipment. Most importantly, the close coupling of the firms combined with the high levels of trust and commitment encouraged extra efforts to continuously improve the situation. This dynamic entrepreneurial element infected even the larger, more bureaucratic partners. The processes in Stages One and Two are summarized in Figure 1.

C. Benefits

The benefits that each side gained once the partnership stage was reached were the result of the evolution of the exchange from the original or primary economic transaction to a

³For elaboration on this notion (usually applied to manufacturing operations) see *Kaizen, The Key to Japan's Competitive Success*, by Masaaki Imai. McGraw-Hill, 1986.

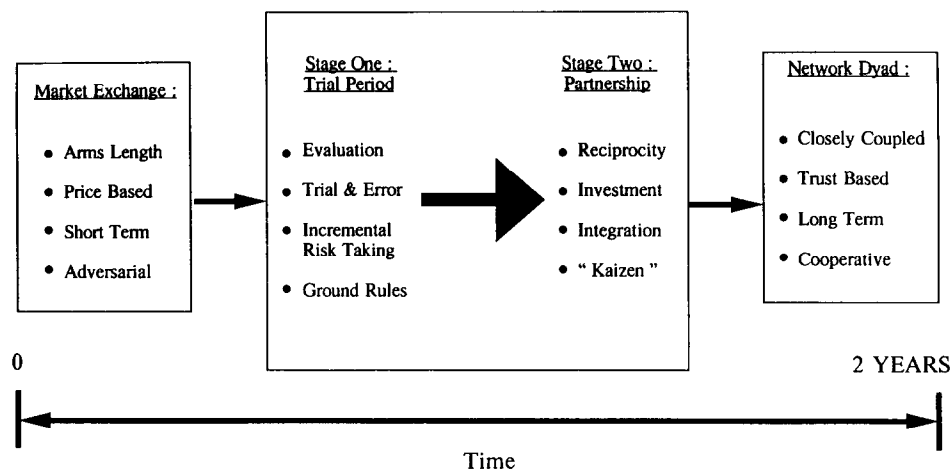


FIGURE 1 Forming the partnership: A two-stage process.

complex web of exchanges connecting the two firms across various levels. The time period for obtaining benefits extended from six months to two years, depending on the complexity of the exchange and the capacities of the players to establish trust and ground rules for cooperation. The sequential evolution is shown in Figure 2.

The conditions created by this expanded exchange process led directly to the following economic benefits for the entrepreneurial companies:

- Access to new channels and markets
- Cost savings through partner's economies of scale
- Shorter lead times for product development
- Technology and process innovations
- Consistent high-quality results
- Extra effort in a crisis
- Market feedback
- Financial resources (e.g., extended payment terms)

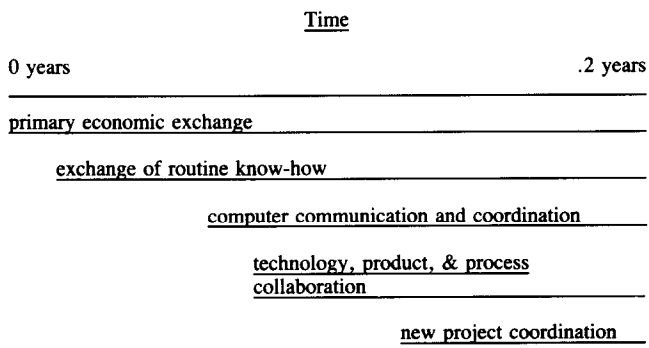


FIGURE 2 The evolution of the partnerships over time (expanding the exchange).

- Access to broad network of information and resources
- Enhanced industry reputation through affiliation.

Benefits to the Other Side

This paper primarily discusses benefits that entrepreneurial firms gained from these partnerships. The other side also received benefits. For the established companies that contracted with the high-growth firms, these transactions represented some of the fastest, if not the fastest, growing sources of new revenues. In one case, the president of the entrepreneurial firm convinced an international printing company to work with his company because of the promise of future, much larger, orders. Each entrepreneurial company had leverage with its larger partners due to this promise of future gains, although some were more aggressive than others in using this point as a selling tool.

The smaller, fast-moving firms also offered larger, more bureaucratic organizations a stimulus for change. A vice president of the large printing company described it in this way:

We are constantly changing things to try to improve the way we are doing business together. We will experiment with new ideas, test new processes, try something different. Relative to the other people we deal with they are much more flexible and willing to experiment with something new. Costs are incurred on both sides but we are willing to pay them. We have learned a lot from them. They have made us a better printing company because they are demanding, innovative, and willing to try things.

Another larger firm described its entrepreneurial partner:

They are more creative than we are. It is the difference between being a victim of your environment or causing the environment to happen.

For this company, the link to the more flexible and innovative smaller firm was a catalyst for the design of a less costly and more efficient joint warehousing and inventory system.

These two benefits—future business growth and a stimulus to process and technology innovation—were common to all of the larger corporate partners. Other benefits varied by alliance but included opening new market segments and shorter product-development lead times. It was clear that a diversity of benefits gained by the more established partners was a strong motivating force for their continued participation in the collaborative ties.

Benefits Through Information Exchange

Many benefits for each side were based on the greater speed, volume, and quality of the information exchanged. A key explanatory factor for the free and rapid movement of information was the non-hierarchical nature of the alliances. The organizations' interactions generally were on a lateral basis, and different layers spoke directly across the boundaries with no, or very few, intermediaries.

The information flows took various forms. The first was the knowledge exchanged that enabled the two companies to routinize the primary economic exchange. This information-exchange process corresponded in time to Stage One of the relationship. This was a period in which, not only was technical information exchanged, but also important inter-organizational and interpersonal learning took place.

The fact that the information exchange is called routine should not diminish its im-

portance. It was through the early interaction driven by economic necessity (getting sufficient know-how transferred quickly to conduct the exchange) and social relations (the incremental growth of trust, norms, and expectations as control mechanisms) that the potential for long-term cooperation was established. In addition, it was only after the foundation of product know-how and procedural routinization was set that knowledge beyond simply learning how to conduct the transaction could begin to be exchanged between the two firms.

The second and perhaps the most important benefit from the long-term alliances was the information exchanged that directly resulted in technology advances and product improvements. The benefits were realized through R&D collaboration; interfirm groups; and mutual assistance in such areas as distribution system design, software development, customized just-in-time inventory systems, and prototype testing. These exchanges typically began after the firms reached Stage Two, the mature, or operations, stage. By this time the firms had learned to combine their resources and technical know-how to move beyond what the organizations could have done individually if their resources alone had been applied. Individuals had established their own linkages with counterparts in the other organizations, information moved quickly, and joint efforts were the norm. These information flows expanded the entrepreneurial organizations' innovative capacities, improved product quality, saved money, and enhanced the companies' market competitiveness. Benefits that came from the computer integration of the firms cut across the first two layers on the diagram because benefits evolved from routine computer use as well as from more extensive exchanges of computerized information. This could range from computerized order entries and order status checks, to more complex and sophisticated integration that resulted in joint modifications of product designs or tightly coordinated scheduling.

In fact, a key to the rapid proliferation of technical benefits was the growing number of computer linkages between alliance partners. Computer linkages were already established in four of the seven alliances and were planned for the remaining ones. These computer ties accelerated the transfer of information and provided technical and administrative feedback loops.

Benefits from Joint Computerized Scheduling and Forecasting

Computer transfer of scheduling and forecasting information was a common occurrence. As an illustration, one manufacturer gave its network of subcontractors a minimum of nine months to a year's planning information about their production needs. This information was transferred directly from a computerized manufacturing resource planning (MRP) system. The expansion of planning information exchanged through computer use yielded greater production efficiencies and improved process controls. The information exchange also helped the supplier schedule its production runs. In addition, feedback from the supplier enabled the manufacturer to improve the effectiveness of its MRP system.

A clothing manufacturing president explained that its alliance with a catalog company could be distinguished from other customer relations by the extent of computerized forecasting information exchanged.

They tell us their needs literally in excess of a year. We appreciate that forecast information and we utilize it for the necessary input toward our capacity planning. We look at how we interface that information with our fabric suppliers and capacity utilization so that we have either operators in position or units to support a particular need. We don't have any customers that give us forecast information that far in advance at all. None.

Benefits from Collaborative R&D

In addition to computer-related benefits, there were numerous other information exchange examples from the alliances that had long-term implications for the firms' innovativeness and competitiveness. A telephone equipment distributor and its manufacturer collaborated routinely on R&D. A telephone distributor's sales and marketing people would collect information from the field, report it back to the manufacturer's design staff, and play an active role in the development of features for new telephone systems and the improvement of old products. According to the distributor's chief operating officer, gains flowed to both sides (in this case both companies were high-growth entrepreneurial firms):

We play an active role with the manufacturer in the development of new products—joint engineering work, R&D collaboration. Their R&D people come up with new features based on the information we provide. Then we field trial their new systems. They trust us to give them feedback and we get to work with state of the art technology.

A heavy manufacturing company that I call Support Products was the most aggressive entrepreneurial company in the formation of alliance relationships to advance product designs. Collaborative technical work was not just encouraged, it was expected with close partners. The chairman of the board talked about the company's joint product-development work with one of its key suppliers:

That's truly been a partnership that has worked both ways. We've gone beyond just being an important customer of theirs to developing technology jointly. We're not aware of any other company that is consistently doing that. The coordinated R&D effort has kept us a jump ahead of the competition.

He went on to describe the benefits of long-term alliances with innovative suppliers:

As long as a relationship is established beyond normal vendor/buyer relationships, then we are privy to new product information from suppliers long before the rest of the world. We get information, technical data, and the first samples.

From the other side of this alliance, a marketing manager talked about the unique joint efforts to develop innovative products and processes:

It's only with Support Products that we have long-term design conceptual engineers talking together; we don't share that information about our planning and design work with the short-term customers. And we haven't agreed to set up a unique just-in-time inventory program with anyone else.

Benefits from Technical Innovation

The long-term commitment between the entrepreneurial computer-hardware company and one of its suppliers resulted in several major cost savings and technological innovations. One technical innovation in particular saved hundreds of thousands of dollars over a six-month period. The supplier changed the grade of wire and recommended modifications to the original electronic connections design. This reduced the "ECO activity." ECO stood for engineering change orders that were retroactive alternations once the printed circuit board was assembled. ECO represented a major source of additional and previously considered unavoidable cost. An unspecified cost had to be added in because even when ECO activity

was corrected (and despite the fact that the boards passed the quality tests), problems still were discovered in the field by the customer. If a board had intermittent failure problems at this point, they were very difficult to correct. The supplier's materials and design change recommendations eliminated this problem.

Other jointly conceived technical innovations had a broad impact on this entrepreneurial firm's final product quality. A direct result was that the "mean time between failure" (MTBF) increased dramatically. The improvement represented an order of magnitude of approximately three times better performance by the company's computer systems. For example, if the MTBF was 30,000 hours before a failure, it increased to 90,000. MTBFs were considered *the* quality measurements for the computer systems. These performance improvements subsequently allowed the entrepreneurial firm to gain new customers whose system uses demanded the higher performance standards.

The president of the clothing manufacturing firm remarked on the importance of collaboration to realize technical and product improvement benefits and the infrequency with which it occurred:

This kind of information exchange beyond the primary product is unusual unless you have mutual commitment and respect. If you have that then it becomes a very common thing. The problem is that you don't reach that with very many people.

It is clear that the benefits that the companies gained from these relationships were not limited to the primary exchange or to direct technical advantage. There were numerous examples of other ripple effects. Technical improvements in one area would transfer into other corporate activities; an innovation in one product would result in upgrades in others. Other benefits included special financing opportunities, shared expenses for mutually beneficial functions (e.g., warehousing), greater outside credibility for the entrepreneurial company through affiliation with established and respected larger firms, and tapping into a larger network of information that the more-established firms provided.

Benefits to the entrepreneurial organization should not be underestimated when a cooperative alliance was established with a more prestigious organization. The reputation (and future business prospects) of a lesser-known organization could be advanced significantly by the existence of a long-term agreement that tied them to a well-known and prestigious firm. The reason this was important was that news of such an alliance gave the smaller firm more legitimacy, since well-established firms are often known to deal only with reputable businesses whose financial strength and product quality are established.

In summary, the alliance exchanges should be viewed as a process that began with a central or primary transaction that ultimately expanded into a diverse collection of exchange relations and benefits. When viewed as a package, these benefits added considerably to the adaptive and innovative capacities for all of the firms, but particularly for the entrepreneurial companies on which this study focused. The examples provided illustrate that the alliances not only generated technical benefits, but that the extent of joint effort and innovative improvements exceeded those in more traditional outside relations. One explanation for this was the ability to make investments in joint technical efforts without having always to justify them with short-term economic results. Under these alliances the longer-term horizon of benefits guided the collaborative efforts around technical advancement. It is also clear that a key element of the alliances that encouraged and accelerated technical improvements was the firms' integration of shared computer systems. These computer linkages held out enormous potential for accelerating the flow of benefits for the entrepreneurial firms.

RISKS AND PARTNERSHIP LONGEVITY

Although this study focused on successful processes of cooperation, these ties had their inherent risks. Alliances offered the advantages of tapping into outside expertise and resources at little or no capital cost. But when firms rely on outside expertise, they fail to develop their own. This vulnerability is accentuated if the firm relies (as some in this sample did) on a single supplier to meet its needs. Heavy dependency on particular individuals for the maintenance of the interfirm relationships is another obvious risk. In some situations this seemed to be an issue, however, in the majority of cases these alliances were maintained through the participation and commitment of numbers of individuals sufficient enough so that the loss of a single person did not threaten alliance viability.

The exchange of technical information highlights a risk inherent in these close collaborative alliances. Often the information exchanged was proprietary. For example, proprietary design information was exchanged, as was proprietary information on new product development and scheduling. The risk always existed that information would be passed on to the competition. In fast-moving markets in which competitive success depended on the rapid introduction of technological innovations, release of proprietary information could have serious economic consequences. Despite the ever-present risk of this happening, there was no evidence that it had happened. It would appear that self-protection (if proprietary information was used inappropriately, the alliance would likely end) and concern for reputation (knowledge in the industry that a firm could not protect confidential information would prevent a firm from gaining new customers) were strong protections against this risk. In addition, by the time these partnerships involved the exchange of proprietary information, each side had committed to a longer-term vision of mutual gain through information sharing. Confidentiality was assumed as each looked not just to immediate gains but, more importantly, to a stream of future returns. Of course, for the smaller entrepreneurial firms with fewer resources on which to fall back, the negative reputational consequences had the potential to be more punishing than for financially stronger and larger organizations.

Shifts in participating firms' strategic interests over an alliance's life are also a threat to the exchange structures. Let us look at what happened in two of the seven cases. By 1987, a five-year-old partnership had ended because the entrepreneurial firm's larger, more bureaucratic partner could not respond to the growing company's changing needs. The partnership was terminated by the smaller firm. In a second case, by 1988, competitive forces in the deregulated telephone industry changed the strategic interests of a partnered firm. An alliance that had lasted five years (and had provided a strong impetus to growth for one entrepreneurial firm) ended when one side violated an exclusive distribution agreement.

As of winter 1990, five alliances were still in place.⁴ Three of these were thriving, the result of increased collaboration and enhanced levels of organizational integration through extensive computer communication. The character of the remaining two partnerships changed, however, when one entrepreneurial company was purchased by a larger corporation in 1988. The new parent company imposed a bottom-line, cost-oriented culture on the purchased firm that by 1990 had replaced the partnership philosophy that had guided the firm's partnerships before 1988. By 1990, trust and loyalty had faded as central aspects of these relationships

⁴By winter 1990, of the four entrepreneurial firms, two had continued their successful growth. In these cases, three of the four partnerships were still intact and flourishing. One high-growth firm that was acquired no longer made public its financial statements. The fourth entrepreneurial firm in the study survived but revenues were low, and new management had to introduce a new product line to turn the company around.

and product quality appeared to be suffering. The advantages that the partnerships had offered, of fast response times, high-quality service, prompt delivery, design cooperation, preferential lead times, and cost discounts, were no longer standard components of the exchange processes in these two cases.

The dissolution stories are not surprising. Market conditions change and strategic interests change. These partnerships cannot be expected to last forever. They are carefully cultivated and maintained structures of exchange, but they have life cycles and obvious risks. Nevertheless, the facts of their creation and dissolution do not contradict the implications of these data that, especially for the entrepreneurial firm, the cultivation of a limited network of such partnerships can contribute significantly to growth and success.

CONCLUSION

It is clear that the firms became tightly integrated through extensive communication and information sharing. Firm representatives made two comparisons: they compared the communication levels with their more typical external relationships; and they compared them with communication that they would expect if the two units were under single ownership (vertically integrated). When the information flows were compared with typical outside business relationships, they were seen as unique and unusual in the lack of bureaucratic tendencies and the effectiveness and openness of communication. Comparable with a unified ownership situation, the firms believed that they were a part of the other company's operations, as though they were a division of the partner organization. Yet the exchange process carried with it none of the bureaucratic requirements and delays associated with functions integrated within large formal organizations.

The close examination of the communication interlinkages revealed joint planning, product development, and technical and process improvements; routinized procedures and systems; computer tie-ins and joint software development to enhance those linkages; informal and routine handling of special requests; and interpersonal relationships based on respect and mutual commitment that further facilitated communication and exchange. Bureaucratic characteristics appeared to be absent from these governance structures. One question is, how can the firms be so closely intertwined with many of the elements found in hierarchies without the bureaucratic drawbacks of typical exchange relations between units within formal organizations?

The explanation that I offer is that the field evidence strongly suggests that these firms have built in the control advantages of smoothly operating, vertically integrated functions while maintaining the flexibility and incentives of market-mediated exchange relations. The first part of the explanation I posit is that the extensive information-sharing and tightly coupled coordination resulted from the realization that mutually competitive and strategic advances were possible through a collaborative effort. In this sense, the organizations could be seen as sharing the same goals, similar to units within vertically integrated firms. Inherent in this situation are powerful controls over opportunistic or complacent behavior. In addition, the history of relations that built the alliances and the personal and organizational commitments and obligations that evolved in that process provided subtle but strong monitoring functions.

The second explanation for these hybrid governance structures is that, despite their integration, the parties are motivated by, and their behavior is checked by, market incentives that remain because the firms are autonomous economic entities. As such they must keep their offerings competitive, or else the partner can and will eventually move to a better

package of goods and services offered by a competitor. Consequently, neither side can take the relationship for granted. They must continue to "sell" the partner on the advantages of the exchange. Commitment must be consistently demonstrated. Extra effort, greater urgency in response to the partner's problems, and expanded information-sharing and joint activities are all logical ways to demonstrate that commitment.

How should entrepreneurial firms choose partners? The evidence suggests that partnering with targeted firms that can provide key resources is a desirable and feasible strategy. A firm should identify prospective partners and evaluate them on their potential to satisfy necessary criteria (e.g., high product quality, delivery, cost, service, and ability to grow with and adapt to the smaller firm's rapid expansion). However, developing cooperative partnerships appears highly dependent on the willingness of each side to move in good faith through a trial period, a process that requires a commitment of trust that not all organizational cultures allow. A prospective partner's track record with other firms can be researched by speaking with their existing partners. Some firms have a history and reputation of forging strong, mutually beneficial alliances, and this information can be gathered at little expense.

The strategy should not be a single-source approach, however. The entrepreneurial firm is too vulnerable to take this risk. Although opportunism would appear to be significantly reduced in partnerships, it is not absent. In addition, unexpected events such as fires or strikes at a partner company could leave a smaller firm in a financially fatal position. One must choose partners carefully and continue to develop a limited set of relationships with others that can stand in for primary ties should crises occur.

Taking these steps should reduce risk of economic disaster resulting from ill-formed partnerships. But obviously, not all risks can be avoided, and not all efforts to establish partnerships will be successful. Yet, a concerted effort to identify and assess prospective partners backward and forward on the value chain—to, in effect, identify a network in which your entrepreneurial firm can become embedded to ensure its success—followed by expressions of strategic interests and low-risk actions that demonstrate that intent will allow each side to test the possibilities.

In summary, it was clear that these linkages contributed to the performance of the firms studied. Note that this research did not attempt to prove causality between successful partnerships and economic performance. High revenue growth is the result of a complex array of market forces, environmental conditions, and organizational decision making. Instead, this study was an exploration of the processes by which the companies maintained these critical cooperative ties. What we can conclude is that each of the four entrepreneurial companies relied on their network of partnerships to meet their performance goals.

Despite this fact, this author knows of no other detailed study of partnership networks built by high-growth smaller firms. In fact, entrepreneurship has not traditionally been defined to extend to the creative cultivation of outside relationships. Looking at the issue from another vantage point, most entrepreneurial firms lack the financial and managerial resources of larger firms and therefore cannot vertically integrate functions to achieve efficiencies. In light of these constraints, these unusually cooperative and long-lasting partnerships represent an alternative organizational strategy by which smaller firms can reap the rewards of vertical integration, while avoiding the financial burdens and bureaucratic inefficiencies of that commitment.

The lasting image from this field study is one of entrepreneurial firms moving in and out of constellations of partnerships as market conditions, strategic interests, and the organizations' cultural capabilities permit. Diagram 2 suggests an alternative way to display

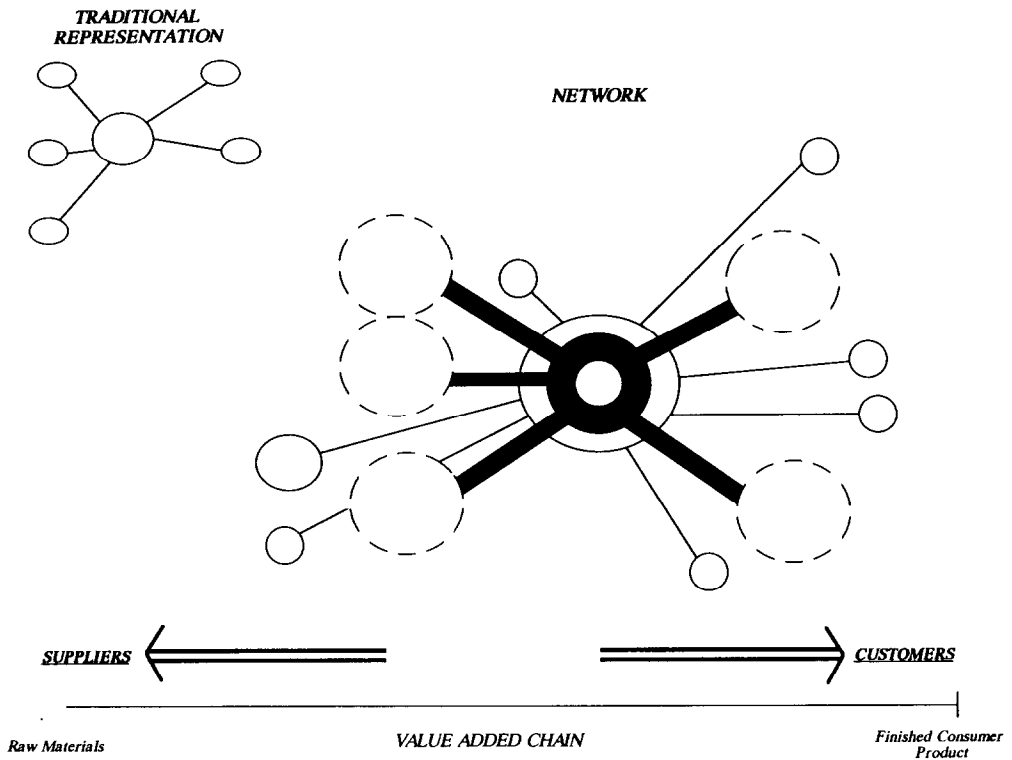


DIAGRAM 2: From Traditional Images to the Network Concept.

these arrangements and compares the new network imagery to the traditional method of illustrating interfirm connections. Although caution must be used in extrapolating from a small sample, the data suggest that carefully selected and managed networks are a strategic necessity for entrepreneurial companies and that this topic needs more research attention.

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